

yousCS 4361 Team 3 Project Proposal

Names: Radhika S, Sameer V, Hudson D, Youssef H

1. Project Overview

Our project is a 3D first-person maze game developed in Java using JOGL (Java OpenGL). The player will navigate through a maze from a first-person perspective using keyboard and mouse controls while searching for an exit. The game will also include collision detection to prevent players from walking through the walls, creating a realistic navigation experience. There will be a timer that tracks how long it takes to complete the maze. It'll show a win screen after you complete it and tell u the time. The program demonstrates fundamental computer graphics concepts such as 3D rendering, camera movement, lighting, user interaction, and collision detection.

2. Goals

- Build a 3D maze that the player can walk through in first-person view
- Add camera and keyboard control to move and look around
- Detect collisions so that the player can't walk through walls
- Add a timer and a win screen when the player reaches the end of the maze

3. Tools/ System/ Software

These are some of the softwares and tools we will be using in our project: Java

JOGL (OpenGL for Java)

Eclipse

GitHub

4. Timeline and Team Responsibilities

Week	[Sameer Vashisth]	[Hudson Duong]	[Youssef Higazy]	[Radhika Sharma]
07/08–07/17 Setup	Set up repo, maze data structure	Basic camera setup	Basic 3D rendering pipeline	Project structure, input handling
07/18–07/27 Core Build	Build maze layout/geometry	Camera movement (WASD/mouse)	Wall/floor models & textures	Player movement logic
07/28–08/05 Gameplay	Maze exit/win logic	Camera-based collision checks	Lighting & visual polish	Collision detection, timer
08/06–08/09 Polish	Integration testing, bug fixes	Integration testing, bug fixes	Integration testing, bug fixes	Integration testing, bug fixes
08/10 Present	Present + demo	Present + demo	Present + demo	Present + demo

08/11–08/12 Report	Write final report	Screenshots & tools section	Design/implementation section	Repo link, lessons learned
-----------------------	--------------------	--------------------------------	----------------------------------	-------------------------------